PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 1 • Chapter OL-T18 Classified

Cross-session • chapter-organised candidate

20 classified items • 55 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 1 • Unit 1 • Chapter OL-T18 • 20 items • 55 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Expanding Brackets

Unit 1 • 55 marks • 20 item(s)

... 5 more item(s) follow

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Working Unit 19

4MA1-2H Q12

ORIGINAL QUESTION
3 marks

12 Expand and simplify $3x(2x+5)(7x-4)$

....

(Total for Question 12 is 3 marks)

Worked Solution 19

Expand three algebraic factors

UNIT 1
3 marks

First two brackets

$$(2x + 5)(7x - 4) = 14x^2 + 27x - 20.$$

Multiply by 3x

$$3x(14x^2 + 27x - 20) = 42x^3 + 81x^2 - 60x.$$

Final answer

$$42x^3 + 81x^2 - 60x$$

Question 02

4MA1-1H Q6

ORIGINAL QUESTION

2 marks

6 (a) Expand and simplify $(m + 5)(m - 8)$

....

(2)

Worked Solution 02

Expand brackets

MODULAR UNIT 1

2 marks

Expand

$$(m+5)(m-8) = m^2 - 3m - 40.$$

Final answer

$$m^2 - 3m - 40$$

Question 28

4MA1-2H Q16 whole

ORIGINAL QUESTION UNIT 1

3 marks

16 Expand and simplify $(2x + 3)(x - 5)(x + 4)$

....

Worked solution 28

Expand three brackets

MODULAR UNIT 1

3 marks

whole – Expand three brackets

Multiply two brackets

$$(x-5)(x+4) = x^2 - x - 20.$$

Multiply by the third bracket

$$(2x+3)(x^2 - x - 20) = 2x^3 - 2x^2 - 40x + 3x^2 - 3x - 60.$$

Collect like terms

$$2x^3 + x^2 - 43x - 60.$$

Final answer

2x^3 + x^2 - 43x - 60

Question 22

4MA1-2H Q17

ORIGINAL QUESTION

3 marks

17 (a) Expand and simplify $(x + 6)(3x - 2)(x + 6)$

....

(3)

Worked Solution 22

Expand three brackets

MODULAR UNIT 1

3 marks

Square the repeated factor

$$(x + 6)^2 = x^2 + 12x + 36.$$

Multiply by the third factor

$$(x^2 + 12x + 36)(3x - 2) = 3x^3 + 34x^2 + 84x - 72.$$

Final answer

$$3x^3 + 34x^2 + 84x - 72$$

Original question 02

4MA1-1H Q3 M02

MODULAR UNIT 1

2 marks

(b) Expand and simplify $2n(4n+3)+n(n-4)$

....

(2)

Worked solution 02

Expand and simplify

MODULAR UNIT 1
2 marks

M02 – Expand and simplify

Expand both products

$$2n(4n+3)+n(n-4)=8n^2+6n+n^2-4n.$$

Collect like terms

$$=9n^2+2n.$$

Final answer

$$9n^2+2n$$

Original question 11

4MA1-1H Q15 M01

MODULAR UNIT 1
3 marks

15 Write $3x(2x-1)(5x+4)$ in the form ax^3+bx^2+cx where a, b and c are integers.

....

Worked solution 11

Expand three factors

MODULAR UNIT 1
3 marks

M01 – Expand three factors

Expand the two brackets

$$(2x-1)(5x+4) = 10x^2 + 3x - 4.$$

Multiply by 3x

$$3x(10x^2 + 3x - 4) = 30x^3 + 9x^2 - 12x.$$

Read a,b,c

$$a = 30,\ b = 9,\ c = -12.$$

Final answer

$$30x^3 + 9x^2 - 12x$$

Original question 02

4MA1-1H Q5 Q5(a)

ORIGINAL QUESTION
2 marks

5 (a) Expand and simplify $(n-6)(n+4)$

....

(2)

Worked solution 02

Chapter: Expanding Brackets

MODULAR UNIT 1

2 marks

Q5(a) – Chapter: Expanding Brackets

Family: Expand two linear expressions • Variant: Standard double bracket

Method

(n - 6)(n + 4) = n^2 + 4n - 6n - 24.

Why: Distribute each term in the first bracket across the second.

Method

Collecting like terms gives n^2 - 2n - 24.

Why: Combine 4n and -6n.

Final answer

expansion: n^2 - 2n - 24

Original question 22

4MA1-2H Q14 Q14(a)

ORIGINAL QUESTION
3 marks

14 (a) Expand and simplify $(5 - x)(2x + 3)(x + 4)$
Show your working clearly.

....

(3)

Worked solution 22

Chapter: Expanding Brackets

MODULAR UNIT 1
3 marks

Q14(a) – Chapter: Expanding Brackets

Family: Expand three or more linear factors • Variant: Three linear brackets

Method
$(5 - x)(2x + 3) = -2x^2 + 7x + 15$.
Why: Expand and collect the first pair of factors.

Method
$(-2x^2 + 7x + 15)(x + 4) = -2x^3 - 8x^2 + 7x^2 + 28x + 15x + 60$.
Why: Distribute every term across the third factor.

Method
The simplified result is $-2x^3 - x^2 + 43x + 60$.
Why: Collect like powers of x.

Final answer

expansion: $-2x^3 - x^2 + 43x + 60$

Original question 10

4MA1-1H Q13 Q13(a)

ORIGINAL QUESTION

3 marks

13 (a) Expand and simplify $5x(x + 2)(3x - 4)$

....

(3)

3

Worked solution 10

Chapter: Expanding Brackets

MODULAR UNIT 1
3 marks

Q13(a) – Chapter: Expanding Brackets

Family: Expand three or more linear factors • Variant: Monomial times two linear brackets

Method
5x(x+2)=5x^2+10x.
Why: Multiply the first two factors.

Method
(5x^2+10x)(3x−4)=15x^3−20x^2+30x^2−40x.
Why: Multiply by the third factor.

Method
Collecting terms gives 15x^3+10x^2−40x.
Why: Combine the two x^2 terms.

Final answer
expanded expression: 15x^3+10x^2−40x

Original question 18

4MA1-2H Q1 Q1(a)

ORIGINAL QUESTION
2 marks

1 (a) Expand and simplify $(y + 4)(2 - y)$

....

(2)

Worked solution 18

Chapter: Expanding Brackets

MODULAR UNIT 1
2 marks

Q1(a) – Chapter: Expanding Brackets

Family: Expand two linear expressions • Variant: Opposite signs, repeated bracket or square

Method
(y+4)(2−y)=2y+8−y^2−4y.
Why: Expand both products.

Method
Collecting terms gives 8−2y−y^2.
Why: Combine the y terms.

Final answer
expanded expression: 8−2y−y^2

7 (a) Expand and simplify $(m - 8)(m + 5)$

....

(2)

Worked solution

January 2020 | Question 7 | 2 marks

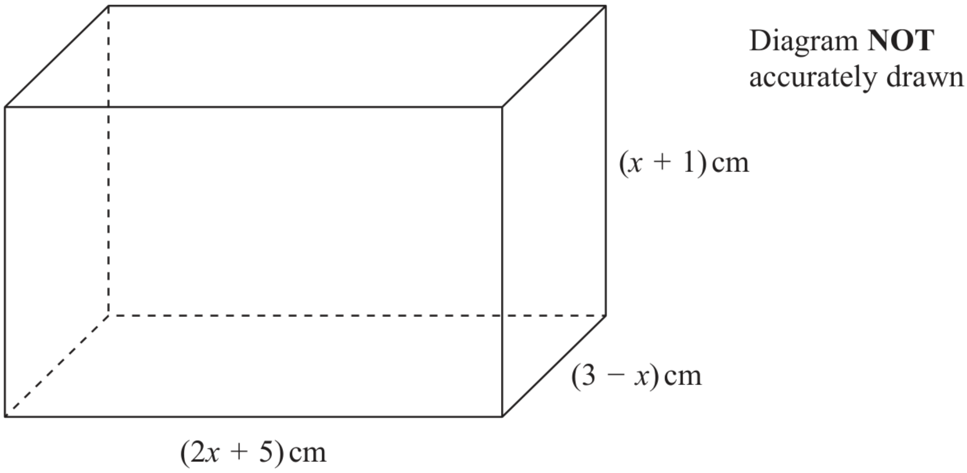
Family: Expand two linear expressions • Variant: Standard double bracket

Method / working

1. Expand: m^2+5m-8m-40.
2. Collect terms to get m^2-3m-40.

Final answer: m^2-3m-40

15



The diagram shows a cuboid of volume V cm^3

(a) Show that $V = 15 + 16x - x^2 - 2x^3$

(3)

Worked solution

January 2020 | Question 15 | 3 marks

Family: Expand three or more linear factors • Variant: Three linear brackets

Method / working

1. $V=(2x+5)(x+1)(3-x)$.
2. $(2x+5)(x+1)=2x^2+7x+5$.
3. Expand with $(3-x)$ to obtain $V=15+16x-x^2-2x^3$.

Final answer: $V=15+16x-x^2-2x^3$

January 2021 | 4MA1-1H | Question 12 | 3 marks

Chapter OL-T18: Expanding Brackets

(b) Expand and simplify $2x(x - 5)(x - 3)$

....
(3)

Worked solution

January 2021 | Question 12 | 3 marks

Family: Expand three or more linear factors • Variant: Monomial times two linear brackets

Method / working

1. First $(x-5)(x-3)=x^2-8x+15$.
2. Multiply by $2x$: $2x^3-16x^2+30x$.
3. Collect terms and state the expansion.

Final answer: $2x^3-16x^2+30x$

January 2021 | 4MA1-2H | Question 5 | 2 marks

Chapter OL-T18: Expanding Brackets

5 (a) Expand and simplify $4x(2x + 5) - 3x(2x - 3)$

....

(2)

Worked solution

January 2021 | Question 5 | 2 marks

Family: Distribute a single term • Variant: Negative term or subtraction of a bracket

Method / working

1. Expand: $4x(2x+5)=8x^2+20x$ and $-3x(2x-3)=-6x^2+9x$.
2. Collect terms to obtain $2x^2+29x$.

Final answer: $2x^2+29x$

June 2021 | 4MA1-1H | Question 5 | 2 marks

Chapter OL-T18: Expanding Brackets

5 (a) Expand and simplify $3x(2x + 3) - x(3x + 5)$

....

(2)

Worked solution

June 2021 | Question 5 | 2 marks

Family: Expand three or more linear factors • Variant: Monomial times two linear brackets

Method / working

1. Expand: $3x(2x+3) - x(3x+5) = 6x^2+9x-3x^2-5x$.
2. Collect like terms to obtain $3x^2 + 4x$.

Final answer: $3x^2 + 4x$

June 2021 | 4MA1-1H | Question 13 | 3 marks

Chapter OL-T18: Expanding Brackets

13 Expand and simplify $4x(3x + 1)(2x - 3)$
Show your working clearly.

....

Worked solution

June 2021 | Question 13 | 3 marks

Family: Expand three or more linear factors • Variant: Monomial times two linear brackets

Method / working

1. Expand two factors, for example $(3x+1)(2x-3)=6x^2-7x-3$.
2. Multiply by $4x$: $24x^3-28x^2-12x$.
3. State the collected expression.

Final answer: $24x^3 - 28x^2 - 12x$

November 2021 | 4MA1-1H | Question 1 | 2 marks

Chapter OL-T18: Expanding Brackets

(b) Expand and simplify $(x - 3)(x + 1)$

....

(2)

Worked solution

November 2021 | Question 1 | 2 marks

Family: Expand two linear expressions • Variant: Standard double bracket

Method / working

1. (b) \((x-3)(x+1)=x^2-2x-3\).

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. Answer: (a) \(e^6\), (b) \(x^2-2x-3\)

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: (b) x^2-2x-3

November 2021 | 4MA1-2H | Question 12 | 3 marks

Chapter OL-T18: Expanding Brackets

(c) Expand and simplify $4x(x - 5)(2x + 3)$
Show your working clearly.

....

(3)

Worked solution

November 2021 | Question 12 | 3 marks

Family: Expand three or more linear factors • Variant: Monomial times two linear brackets

Method / working

1. (c)

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. $4x(x-5)(2x+3)=4x(2x^2-7x-15)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. $=8x^3-28x^2-60x$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. Answer: (a) $16p^6q^8$, (b) $\frac{17}{30x}$, (c) $8x^3-28x^2-60x$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: (c) $8x^3-28x^2-60x$

January 2023 | 1H | Question 13 | 3 marks

Chapter OL-T18: Expanding Brackets

13 Expand and simplify $3x(2x-5)^2$

Show clear algebraic working.

....

(Total for Question 13 is 3 marks)

Answer reference | January 2023 | Question 13

Family: Distribute a single term • Variant: Positive monomial over a bracket

13	$3x(2x-5)=6x^2-15x$ **or**		3	M1 for multiplying $3x$ by $(2x-5)$ with **both** terms correct or
	$(2x-5)^2=4x^2-10x-10x+25$ **or**			for multiplying $(2x-5)$ by $(2x-5)$ with 3 out of 4 terms correct or
	$(2x-5)^2=4x^2-20x+25$			for multiplying $(2x-5)$ by $(2x-5)$ and getting $4x^2-20x\ldots$ **or** $\ldots-20x+25$ (not for $4x^2+25$)
	$(6x^2-15x)(2x-5)=12x^3-30x^2-30x^2+75x$ oe **or**			M1ft (dep) for multiplying the product of $3x$ and $(2x-5)$ by $(2x-5)$ with 3 out of 4 terms correct or
	$(6x^2-15x)(2x-5)=12x^3-60x^2+75x$ oe **or**			for multiplying the product of $3x$ and $(2x-5)$ by $(2x-5)$ and getting $12x^3-60x^2\ldots$ **or** $\ldots-60x^2+75x$
	$3x(4x^2-10x-10x+25)=12x^3-30x^2-30x^2+75x$ oe **or**			for multiplying the product of $(2x-5)$ and $(2x-5)$ by $3x$ with 3 out of 4 terms correct or
	$3x(4x^2-20x+25)=12x^3-60x^2+75x$			for multiplying the product of $(2x-5)$ and $(2x-5)$ by $3x$ with 2 out of 3 terms correct or
				Expansion in one stage will lead to $12x^3-30x^2-30x^2+75x$ without firstly expanding two factors – award M2 for 3 out of 4 terms correct M1 for 2 out of 4 terms correct
	Working required	$12x^3-60x^2+75x$		A1 dep on M1
				Total 3 marks

Worked solution

January 2023 | Question 13 | 3 marks

Family: Distribute a single term • Variant: Positive monomial over a bracket

Official final mark-scheme pages are embedded immediately after this question.

14 (a) Expand and simplify $7x(3x + 2)(2x - 5)$

....

(3)

(b) Solve $\dfrac{9}{2y} + \dfrac{5}{7} = 5$

Show clear algebraic working.

$y =$ ....

(3)

(Total for Question 14 is 6 marks)

Worked solution

November 2025 | Question 14 | 6 marks

Family: Distribute a single term • Variant: Positive monomial over a bracket

Solution

(a)

$7x(3x+2)(2x-5)=7x(6x^2-11x-10)$

$=42x^3-77x^2-70x$

(b)

$\frac{9}{2y}+\frac{5}{7}=5$

$\frac{9}{2y}=\frac{30}{7}$

$63=60y$

Answer: (a) $42x^3-77x^2-70x$, (b) $y=\frac{21}{20}$

PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION